DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

University Examinations 2020/2021

SECOND YEAR SEMESTER II EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN COMPUTER SCIENCE

CCS 2208: COMPUTER GRAPHICS

DATE: SEPTEMBER 2020 **TIME: 2 HOURS**

Instructions: Answer Question 1 and Any Other Two.

Question 1: (30 Marks)

a) Suppose RGB raster system is to be designed using an 8 inch X 10 inch screen with a resolution of 100 pixels per inch in each direction. If we want to store 8 bits per pixel in the frame buffer, how much storage (in kilobytes) do we need for frame buffer? **(3 marks)**

b) Describe three uses of chromaticity diagram **(3 marks)**

c) Distinguish between the following terms
   i. Emissive and non-emissive displays **(2 marks)**
   ii. Image processing and image synthesis **(2 marks)**
   iii. Interlacing scan and progressive scan **(2 marks)**

d) The Phongs reflection model is a sum of three terms that each models a specific type of light reflection. Name and describe the three terms. **(3 marks)**

e) Discuss the architecture of a virtual reality system. **(4 marks)**

f) Njuguna wants to acquire a graphics display device, explain four factors she needs to consider in the acquisition process. **(4 marks)**

g) Scale a triangle with vertices at original coordinates (10, 25, 5), (5, 10, 5), (20,10,10) by S_x=1.5, S_y=2, and S_z=0.5 with respect to the origin. **(3 marks)**

h) Suppose you have an image which you want to store in permanent storage, describe four factors you would consider when selecting a file format. **(4 marks)**

Question 2: (15 Marks)

a) Describe one technique used in hidden surface removal. **(1 mark)**

b) The reflection along the line y=x is equivalent to the reflection about the X axis followed by counter clockwise rotation by θ degrees. Find the value of θ. **(3 marks)**

c) Rasterize a line having end points (10,20) and (16,25) and plot the resultant pixels using Bresenham's algorithm. **(5 marks)**

d) Suppose you are given a scene to render through the graphics pipeline. Explain the Cartesian reference frames and procedure you would use to construct and display the scene. **(6 marks)**

Question 3: (15 Marks)

a) Briefly discuss the purpose of the following methods used in creating 3-D sceneries
   i. Parallel projection **(2 marks)**
   ii. Perspective projection **(2 marks)**

b) Given 3D geometric primitives explain how you would generate an image using the 3D rendering pipeline. **(6 marks)**

c) Write a simple OpenGL program for rendering on a black background a red square. **(5 marks)**

Question 4: (15 Marks)

a) Given a light source, describe two factors that will affect the appearance of an object.**(2 marks)**

b) Consider two raster systems with the resolutions of 1280 x 720 and 2048 x 1536.
   i. Describe two ways of refreshing a display screen. **(2 marks)**
   ii. How many pixels could be accessed per second in each of these systems by a display controller with a refresh rate of 120Hz? **(2 marks)**
   iii. What is the access time per pixel in each system? **(2 marks)**

c) Film animators use a variety of methods for depicting and emphasizing motion sequences. Discuss two traditional animation techniques giving examples. **(4 marks)**

d) How is colour represented in computer graphics and how does this relate to the Human Visual System. **(3 marks)**